

Making Waves with Fourier Series

Introductory Digital Signal Processing

Teacher Notes

Level: Introductory

Objective: A student will have a basic idea of how Fourier Series work in describing waves.

Pre-Knowledge of Students: Basic understanding of waves

Key Terms: Fourier, waves, sine, cosine, amplitude, frequency, wavelength

Related Videos for Teacher: None

Teacher Preparation:

Most likely it will be difficult to incorporate GNU Radio and Fourier Series into the high school science class and curriculum. This difficulty is partly due to the learning curve for GNU Radio and partly due to the high level of math understanding needed to understand Fourier Series. However, the main concept behind Fourier Series: any wave can be recreated with the sum of sine and cosine waves, can be understood by our high school students (one can also think of the concept of superposition). An interactive module has been created in GNU Radio where our students can explore adding different sin and cosine waves. One can imagine having this module running and having one or a group of students create waves. The first lesson could just be guided instruction by having the students examine the results of playing with certain sliders and combinations of sliders. After that the student could be asked to make a square, triangle, or sawtooth wave. The following is an example lesson plan.

What is Needed:

Enough computers for the class where students work alone or in pairs (computers must also be running LINUX (UBUNTU is the most popular version))

A projector attached to your computer and a laser pointer

GNU Radio (software program) and `Fourier_Series_Wave_Explorer.grc`

As a teacher you should be familiar with GNU Radio and running a LINUX computer.

You should run through the exercise and set up your student's computers ahead of time. Your students do not need to know LINUX or how to operate GNU Radio. If the program hangs up, you can show them how to stop and restart the program.

Activity Notes:

Introduction:

- a. Discuss the concept of any complex wave can be created by adding together simpler single frequency sine and cosine waves (basic idea behind Fourier Series and superposition).
- b. Have the students now turn on or look at the program on their computer (you should already be running GNURadio). You can talk a little bit about GNURadio and what one can do with the program.

Guided Instruction:

- a. Explain what the students are seeing and the function of each graph, slider, and toggle.

- b. Adjust all of the sliders (one at a time, positive 1 and negative 1 directions) and let the students see what happens to the wave in the Sum of Waves/Mystery Wave Graph (Mystery Wave is turned off). You can walk around the class and check that the students are playing with the different sliders/toggles.
- c. Now have the students focus their attention on the Wave 1 graph. Ask them to count the number of waves present for Wave 1. Show them how to count a wave (peak to peak, trough to trough, or same starting point to same starting point). The students should see eight complete waves. Have the students count the number of waves shown for Waves 2 through 5. Ask them if they notice a pattern. The students should see that each graph has eight more waves than the previous graph. The students may also say that each successive graph shows a multiple of the first graph. You can introduce the word harmonic at this point.

Self-Discovery:

- a. After seeing that the students have played with each of the sliders and created a complex wave, have the students turn on the mystery wave by moving the Mystery Wave slider left or right (toward the negative 1 or positive 1 direction). The toggle default is the square wave. Ask the students to try and match the square wave by adjusting any or all of the toggles for waves 1 through 5. Walk around the class and provide hints as necessary. Compliment the students who are doing a good job. You should allow 10 minutes for this section of the activity.
- b. If most of the class is successful in making a square wave, have the students change the toggle to a sawtooth. Again, walk around the class and provide hints as necessary. Again, compliment the students who are doing a good job. Again, allow 10 minutes for this section of the activity.
- c. Finally, if most of the class is successful in making a square wave, have the students change the toggle to a triangle wave. The process is the same as in part a and b.

Wrap-Up:

- a. If the lesson is running long part c may be omitted.
- b. Again, compliment the students for their hard work and successes. Explain to them that what they did by experimenting can also be found mathematically (Fourier Series).